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COURSE NAME: CSA0905RA JAVA PROGRAMMING

CO3 AT1 – ASSIGNMENT



Secure Exit

SIMATS Online Programming Lab
JAVA PROGRAMMING

 SARAN S
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10 marks

Q4
Online Food Delivery – Multiple Threads and Execution Order
10 marks

Q5
Manufacturing Plant – Producer-Consumer Problem
10 marks

4/5 answered

Your Code

```
1 import java.util.*;
2 class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         try{
6             String name=sc.next();
7             int age=sc.nextInt();
8             double temperature=sc.nextDouble();
9             if(age<18||age>100){
10                throw new InvalidAgeException("Invalid age");
11            }
12            if(temperature<95.0 || temperature>104.0){
13                throw new InvalidTemperatureException("Invalid temperature");
14            }
15            System.out.println("Patient registered successfully");
16        }
17        catch(InvalidAgeException e){
18            System.out.println("InvalidAgeException");
19        }
20        catch(InvalidTemperatureException e){
21            System.out.println("InvalidTemperatureException");
22        }
23        finally{
24            System.out.println("registration completed.");
25        }
26    }
27 }
28 class InvalidAgeException extends Exception{
29     InvalidAgeException(String message){
30         super(message);
31     }
32 }
```

Input

Arun 35 98.6

Output

Patient registered successfully
registration completed.



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Input / Scenario Expected Result Purpose

1, 2 Valid product; total cost calculated Normal case
5, 2 ArrayIndexOutOfBoundsException handled Boundary / exception case
2, -1 Invalid quantity Invalid input
abc, 2 InputMismatchException handled Invalid input

Your Code

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         String[] product={"Laptop","Mobile","Headphones"};
6         double[] price={50000,20000,2000};
7         try{
8             int index=sc.nextInt();
9             int quantity=sc.nextInt();
10            if(quantity<=0)
11                throw new IllegalArgumentException("Invalid quantity");
12            System.out.println(" Product: "+product[index]);
13            System.out.println(" Total cost: "+(price[index]*quantity));
14        }catch(ArrayIndexOutOfBoundsException e){
15            System.out.println("ArrayIndexOutOfBoundsException handled");
16        }catch(InputMismatchException e){
17            System.out.println("InputMismatchException handled");
18        }catch(IllegalArgumentException e){
19            System.out.println(e.getMessage());
20        }
21    }
22 }
```

Input

1 2

Output

Product: Mobile
Total cost:40000.0

Submitted



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4/5 answered

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args) throws Exception{
4         Scanner sc=new Scanner(System.in);
5         String order=sc.next();
6         String payment=sc.next();
7         String delivery=sc.next();
8
9         orderThread t1=new orderThread();
10        PaymentThread t2=new PaymentThread(payment);
11        DeliveryThread t3=new DeliveryThread(payment);
12        t1.start();
13        t1.join();
14        t2.start();
15        t2.join();
16        t3.start();
17        t3.join();
18    }
19 }
20 class orderThread extends Thread{
21     public void run(){
22         System.out.println("order processing completed");
23     }
24 }
25 class PaymentThread extends Thread{
26     String payment;
27     PaymentThread(String payment){
28         this.payment=payment;
29     }
30     public void run(){
31         if(payment.equalsIgnoreCase("Failed"))
32             System.out.println("Delivery assignment cancelled");
33         else
34             System.out.println("Delivery assignment completed");
35     }
36 }
```

Pizza UPI AgentA

Output



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4/5 answered

Personalizing menu
Producer-Consumer Problem
10 marks

Your Code

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args) throws Exception{
4         Buffer b=new Buffer();
5         Thread producer=new Thread(){
6             public void run(){
7                 try{
8                     b.produce("P1");
9                     b.produce("P2");
10                    b.produce("P3");
11                }catch(Exception e){
12                }
13            }
14        };
15        Thread consumer=new Thread(){
16            public void run(){
17                try{
18                    b.getProduct();
19                    b.getProduct();
20                    b.getProduct();
21                }catch(Exception e){
22                }
23            }
24        };
25        producer.start();
26        consumer.start();
27
28        producer.join();
29        consumer.join();
30    }
31 }
32 class Buffer{
33     String products;
```

Input

stdin input

Output

product: P1
consumed: P1
product: P2
consumed: P2
product: P3
consumed: P3

```
15  Thread consumer=new Thread(){
16      public void run(){
17          try{
18              b.getProduct();
19              b.getProduct();
20              b.getProduct();
21          }catch(Exception e){
22              }
23          }
24      };
25      producer.start();
26      consumer.start();
27
28      producer.join();
29      consumer.join();
30  }
31  }
32  class Buffer{
33      String product;
34      boolean full=false;
35      synchronized void produce(String p) throws Exception{
36          while(full)
37              wait();
38          product=p;
39          full=true;
40          System.out.println("product: "+product);
41          notify();
42      }
43      synchronized void getProduct() throws Exception{
44          while(!full)
45              wait();
46          System.out.println("Consumed: "+product);
47          full=false;
48          notify();
49      }
50  }
```